

Name: Yihang Yu

Primarily, Git and other familiar version control tools offer a comprehensive history of all code changes, allowing developers to track every modification, understand its purpose, and identify who made it. This detailed record is crucial for debugging and auditing. Secondly, version control tools enable seamless collaboration among team members. Developers can work on different sections concurrently through branching, and then efficiently integrate their contributions without repeating others' parts. This minimizes conflicts, unnecessary repetition and significantly boosts team productivity. Moreover, the ability to easily revert to any previous stable version of the codebase plays as a critical safety net, which could prevent data loss and promote errors recovery. Ultimately, version control tools strengthen code quality as well as provides a robust mechanism for managing complex projects.